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Original Article

Developing approaches to incorporate donor-lung computed tomography images into machine learning models to predict severe primary graft dysfunction after lung transplantation

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ABSTRACT

Primary graft dysfunction (PGD) is a common complication after lung transplantation associated with poor outcomes. Although risk factors have been identified, the complex interactions between clinical variables affecting PGD risk are not well understood, which can complicate decisions about donor-lung acceptance. Previously, we developed a machine learning model to predict grade 3 PGD using donor and recipient electronic health record data, but it lacked granular information from donor-lung computed tomography (CT) scans, which are routinely assessed during offer review. In this study, we used a gated approach to determine optimal methods for analyzing donor-lung CT scans among patients receiving first-time, bilateral lung transplants at a single center over 10 years. We assessed

Abbreviations: AUPRC, area under the precision-recall curve; AUROC, area under the receiver operating characteristic curve; CT, computed tomography; DICOM, Digital Imaging and Communications in Medicine; EHR, electronic health record; LAS, lung allocation score; LASSO, least absolute shrinkage and selection operator; ML, machine learning; NPV, negative predictive value; PGD, primary graft dysfunction; PPV, positive predictive value.

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4 computer vision approaches and fused the best with electronic health record data at 3 points in the machine learning process. A total of 160 patients had donor-lung CT scans for analysis. The best imaging-only approach employed a 3D ResNet model, yielding median (interquartile range) areas under the receiver operating characteristic and precision-recall curves of 0.63 (0.49-0.72) and 0.48 (0.35-0.6), respectively. Combining imaging with clinical data using late fusion provided the highest performance, with median areas under the receiver operating characteristic and precision-recall curves of 0.74 (0.59-0.85) and 0.61 (0.47-0.72), respectively.

1. Introduction

Outcomes after lung transplantation remain disappointing, with a median survival of 6.7 years.[1] Primary graft dysfunction (PGD) is an acute lung injury occurring in the first 72 hours after transplantation. PGD is defined by the presence of pulmonary edema on chest x-ray and impairment in gas exchange.[2] Severe PGD (grade 3) occurs in approximately 30% of recipients and is associated with poor short- and long-term outcomes including duration of mechanical ventilation, hospital length of stay, development of chronic lung allograft dysfunction, and survival.[3-7] PGD is believed to be the result of multiple insults that begin in the donor with brain death, ventilator-induced lung injury, and aspiration, followed by cold ischemia after organ recovery then reperfusion at the time of transplantation.[8] Donor, recipient, and surgical risk factors for PGD have previously been identified.[3,9] However, how these risk factors interact and how clinicians should consider these in making decisions about donor organ acceptance for a given recipient remain unclear. Currently, lungs are accepted from <30% of organ donors nationwide, and >50% of accepted donors are considered extended criteria donors.[10,11] The risk of PGD impacts the decision to accept a donor offer for a given recipient. In a survey of pulmonologists and surgeons, 75% of respondents believed that a model predicting the risk of PGD would influence the decision to proceed with a transplant, and 94% thought that a predictive model would influence the surgical plan.[12]

Several donor score calculators have been developed to predict acceptability for transplantation and posttransplant outcomes.[10,13-15] However, none of these has been widely adopted because of inconsistent results and suboptimal performance. Clinicians estimate the potential risk of PGD based on donor gas exchange, donor-lung computed tomography (CT) scan findings, and recipient characteristics to decide about organ acceptance. However, these estimations are imprecise and subjective and may contribute to the underutilization of donor lungs. To enhance clinical decision making about donor acceptance, we previously developed a machine learning (ML) model to predict severe PGD based on donor and recipient characteristics.[16] Although the model was reasonably predictive, it did not include donor-lung CT images, a cornerstone of clinical decision making. We hypothesize that including donor-lung CT image features in a predictive model would improve model performance. However, the optimal approach to incorporating image-derived features with corresponding clinical phenotype data in a predictive model is unclear. Therefore, this study aimed to determine if donor-lung CT image features could predict the development of severe PGD and to define the optimal method for encoding and fusing the 3D feature-space of CT scans with clinical phenotyping data to predict severe PGD.

2. Methods

2.1. Study population

All patients aged ≥18 years who underwent bilateral lung transplantation at Washington University in St. Louis between January 1, 2010 and December 31, 2019 and agreed to participate in our lung transplant registry and had donors with lung CT scans available for analysis were included in the study. Patients were excluded if they underwent simultaneous multiorgan transplantation or a single lung transplant. For patients who underwent retransplantation, only their first transplant was included. No specific study protocol was prepared. This study was approved by the Washington University in St. Louis institutional review board (#202012155).

2.2. Data collection

All available donor CT scans were collected from Mid-America Transplant and other referring organ procurement organizations for study inclusion; however, only CT scans for accepted and transplanted donor lungs were used for this analysis. All CT scans were performed by referring centers according to local imaging protocols. Despite differences in acquisition technique, all available images were used, which enhanced model robustness and generalizability.

Clinical phenotype data from the United Network for Organ Sharing was combined with information hand-extracted from the donor portal and local electronic health record (EHR) and stored in a Health Insurance Portability and Accountability Act-secure institutional instance the REDCap electronic data capture platform.[16-18]

2.3. Outcome

The primary outcome was severe PGD (grade 3) at any time within the first 72 hours after transplantation as defined by

International Society for Heart and Lung Transplantation criteria.[2,16] PGD was graded by 2 board-certified pulmonologists (A.P.M., R.H.) as previously reported.[16]

2.4. Model development

Overview: Model development was conducted in a gated, 3-step process and follows the Tripod + AI reporting standard (Supplementary Fig. 1). First, we identified the optimal method for predicting severe PGD using donor-lung CT scan images through transfer learning, integrating approaches successful in lung pathology identification.[19-30] Second, we aggregated clinical phenotyping data from the local and donor EHRs and performed a feature ablation study to reduce the number of utilized data points to help prevent model overfitting.[16] Third, we evaluated the predictive performance of different methods for fusing the clinical and imaging data utilizing the best method from step 1 and the selected clinical phenotype elements from step 2.[31]

In the first step, for the optimal method for predicting severe PGD using only donor-lung CT scans, 4 modeling approaches were assessed (Fig. 1):

Approach 1 (random montage): In this approach, the thorax was divided into 10 sections, and 1 slice from each section was randomly selected without replacement and placed into a 1 × 10 montage in random order.[19,20] The montages were then fed into a ResNet50 model for training. A ResNet model was selected for this approach based on its strong performance in clinical predictive modeling.[19-25]

Approach 2 (sequential montage): This approach uses the same methodology as Approach 1; however, the slices are ordered sequentially.[19-25]

Approach 3 (slice-based feature extraction): In this approach, the thorax was divided into 10 sections, and 10 slices were randomly selected as above. This methodology extracts features from each CT scan slice from the pretrained EfficientNet B7 model.[26-28] The resultant, ordered, feature vectors were concatenated together and then fed through a single fully convolutional network layer, whose weights were updated during training.

Approach 4 (3D modeling): Each CT scan was divided into 10 sections and placed into a 1 × 10 montage, as in Approach 2.[29,30] The resultant montage was then passed into a pretrained 3D ResNet model (SE ResNet50 3D). The resultant vectors were then passed through a single fully convolutional network layer with weights updated during training.

All models were developed using Digital Imaging and Communications in Medicine-encoded images. For approaches 1 and 2, images were scaled to a resolution of 512 × 512 and center cropped to 380× 380. Because of computational limitations, the images were further scaled to 244 × 244 for approaches 3 and 4. The images were then grayscale normalized, and the top 15% and bottom 5% of each CT scan series were automatically and empirically discarded to remove extrathoracic structures before slices were selected for further processing.[28] This step enriched the CT scans for thoracic structures and reduced compute-resource requirements. For each modeling approach, the cohort was divided into a stratified 75%:12.5%:12.5% train:test:validation set, such that 12.5% of the CT scans were never seen by the model during development. During training, 7-fold cross validation was performed with the best-performing model by area under the precision-recall curve (AUPRC) applied to the held-out validation set. To measure the degree of error, a bootstrap analysis of 100 iterations was performed for each approach. Each model had a learning rate of 1×10^{-5}, and the model weights were optimized using a focal loss function set to maximize the AUPRC. Hyperparameters were selected during cross validation, and gamma values were tested in intervals of 0.5 ranging from 0 to 3. A terminal SoftMax function was used to determine outcome probabilities. All models were trained on the institutional high-performance compute cluster, using one Nvidia A100 graphics processing unit.

For the second step, data from the United Network for Organ Sharing data repository and combined with local EHR data (16). Although present in the United Network for Organ Sharing data, the lung allocation score (LAS) was not included in the modeling process because it is no longer routinely used in the clinical setting. This decision was made to enable the creation of a model that closely resembles the clinical decision-making process by utilizing only readily available data. However, a sensitivity analysis in which the LAS feature remained was also performed to assess for degraded model performance. A least absolute shrinkage and selection operator (LASSO) regression was performed on these data using Akaike's information criterion to reduce the number of clinical features used in the prediction model. To select features that were most consistently identified as important, a LASSO regression was performed 100 times, and the selected features from all iterations were tallied.[16] Features selected in ≥10% of the iterations were used in the final prediction model (step 3).

In the third step, the optimal method for fusing multimodal data (imaging and EHR-derived clinical phenotyping data) was assessed by combining the best-performing method from step 1 and the LASSO-selected features from step 2.[16] In this step, these multimodal data were fused together at different stages in the ML pipeline and the loss was backpropagated at the indicated stages (Fig. 2).[31] As a baseline, only EHR data selected by the LASSO regression was used to predict PGD using 2 fully convolutional network layers.

2.5. Statistical analysis

Model performance was assessed using area under the receiver operating characteristic curve (AUROC), AUPRC, positive predictive value (PPV), negative predictive value (NPV), accuracy, and F1 score from the aggregated performance on the held-out test set from all bootstrap iterations. Performance metrics are reported as median (interquartile range). Model probabilities >0.5 were considered positive for PGD. Missing clinical data was median imputed (Supplementary Information, Missing Data). Model development was performed in Python 3.8 (Python Software Foundation) using Jupyter Notebook (NumFOCUS) and Pytorch (version 1.13.1).[32]

Between-group comparisons were performed using the Mann–Whitney U test for continuous data and Fisher exact test

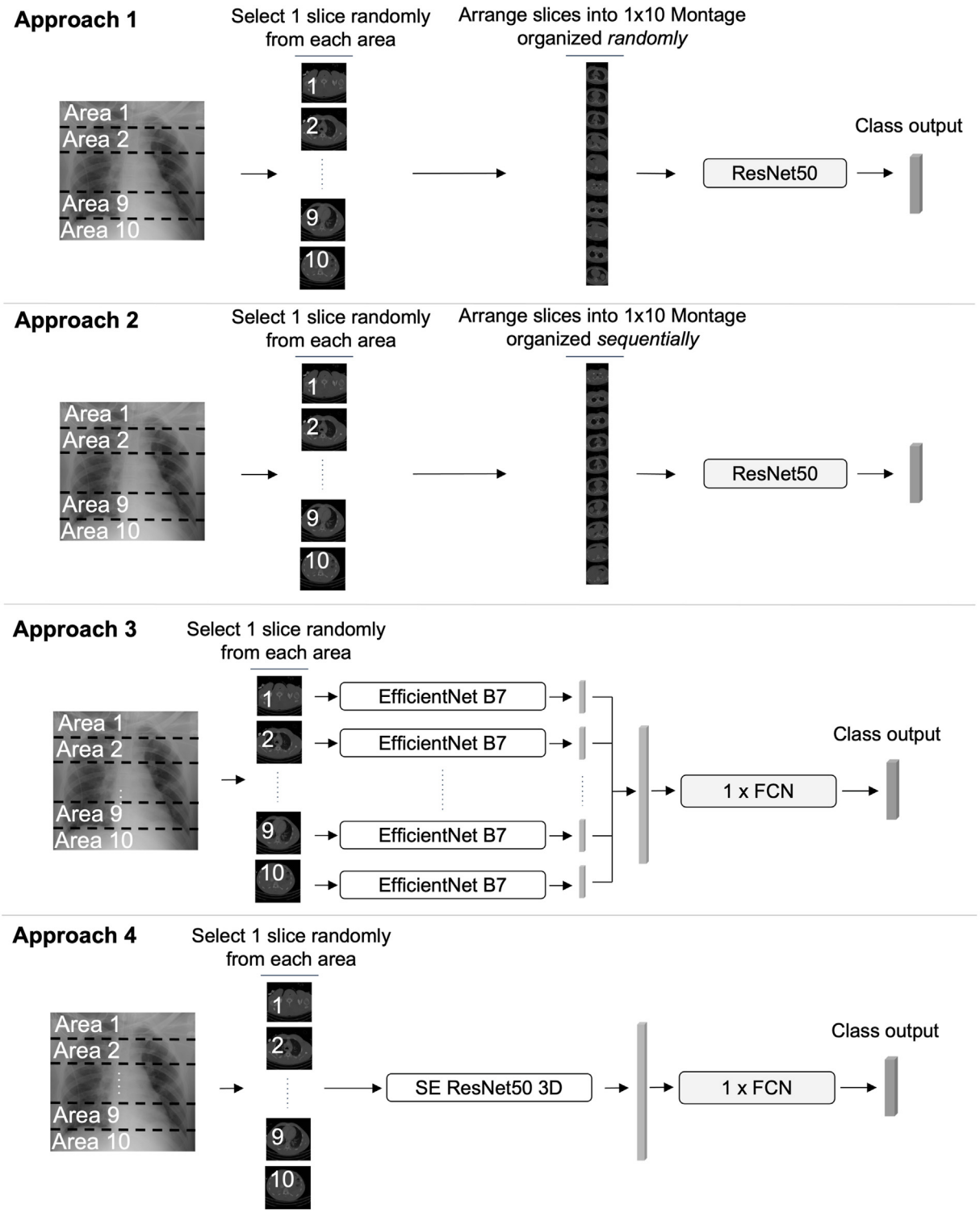

Figure 1. Modeling architectures. This schematic outlines the methodologies applied to ascertain the most effective strategy for predicting grade 3 PGD using donor-lung CT scans. In each approach, the top 15% and bottom 5% of each CT scan series class were removed. Training was performed using a stratified, 7-fold cross validation at a ratio of 75%:12.5%:12.5% for training:validation:testing with a learning rate set to 1×10^{-5}, using focal loss set to maximize area under the precision-recall curve with a terminal SoftMax activation function. **Approach 1**: This method involves partitioning the lung into 10 distinct regions. From each region, a single CT scan slice is randomly selected. These slices are then assembled into a 1×10 montage, arranged in a random sequence. This montage is subsequently processed through a ResNet50 convolutional neural network. **Approach 2**: Similar to Approach 1, but the images are sequentially arranged preserving the order of capture. **Approach 3**: The donor-lung CT scan is segmented into 10 sections. One slice from each section is randomly chosen and analyzed by the EfficientNetB7 model for feature extraction. The resulting feature matrices from each slice are concatenated and subsequently fed into a fully convolutional network layer. **Approach 4**: Similar to Approach 2, but the images are processed through a 3D ResNet model. CT, computed tomography; FCN, fully convolutional network; PGD, primary graft dysfunction.

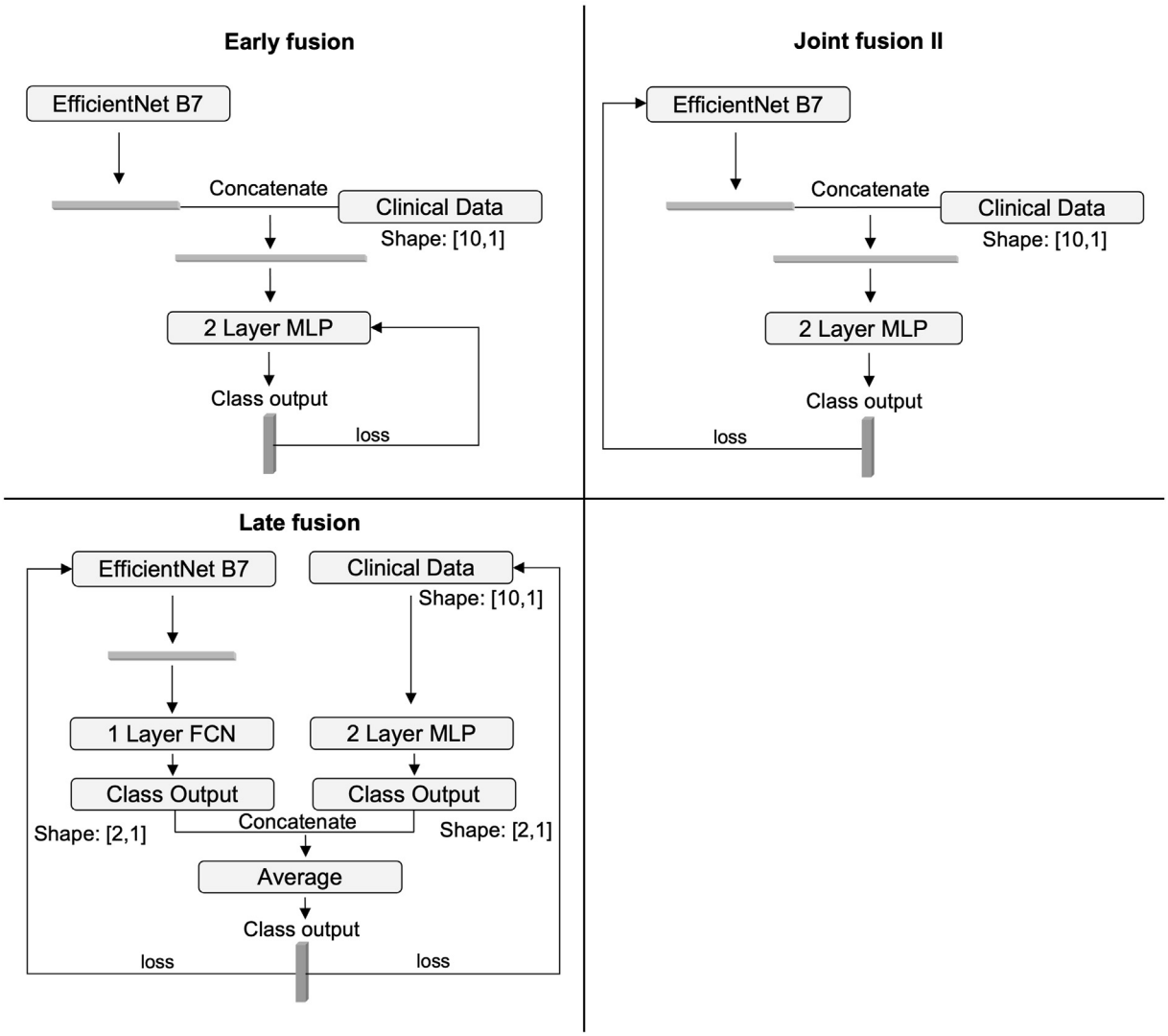

Figure 2. Discrete data and donor-lung CT scan fusion architecture. Schematic representing the 3 different fusion points for the discrete clinical and CT scan data (early, joint, and late). All approaches used the EfficientNet B7 architecture for fusion of imaging data and the SoftMax function for categorization. **Early Fusion**: Feature extraction is performed using the EfficientNet B7 model with frozen weights. The resultant vector is concatenated with clinical data and passed to an MLP. This MLP has its weights updated during the training process before final categorization is determined. **Joint Fusion**: Similar to early fusion, however, the loss is backpropagated through both the pretrained EfficientNet B7 and the MLP. **Late Fusion**: In this variant, both imaging and clinical data are initially processed independently through separate MLPs. Post-processing, the class probabilities derived from each MLP are concatenated and averaged. Here, the loss is backpropagated to the EfficientNet B7 model and the downstream MLPs. CT, computed tomography. FCN, fully convolutional network; MLP, multilayer perceptron.

for categorical data with R Version 4.1.2 (R Foundation for Statistical Computing) using RStudio (RStudio, PBC). Between model comparisons were assessed using the Kruskal–Wallis test. A P value $< .05$ was considered statistically significant.

3. Results

3.1. Study population

In total, 160 patients were eligible for inclusion, and severe PGD within 72 hours of transplant occurred in 44 (27.5%) patients (Table 1, Supplementary Table 1). Patients who developed severe PGD, were more likely to have a higher median LAS (37.6 [34.74-44.42] vs 36.05 [33.36-40.13]; $P = .01$), a primary diagnosis of interstitial lung disease (65.91% vs 41.24%; $P = .01$), and less likely to have chronic obstructive pulmonary disease (6 [13.64%)] vs 46 [39.66%]; $P = .001$). A greater proportion of patients who developed severe PGD were hospitalized at the time of transplant (34.09%) compared to those who did not develop severe PGD (20.69%), but this was not statistically significant ($P = .099$). There was no significant difference in the proportion of patients who developed severe PGD who required intensive care unit admission, mechanical ventilation, or extracorporeal membrane oxygenation support before transplant and those who did not develop severe PGD, but this analysis was limited by a small sample size (Table 1). As it became standard practice to obtain donor-lung CT scans over the study period, a comparative analysis between those with and without imaging

Table 1
Population characteristics.

Characteristic	All (n = 160)	PGD (n = 44)	No PGD (n = 116)	*P* value (PGD vs no PGD)
Recipient				
Age, y, median (IQR)	61 (52-66)	60.5 (48-67)	61 (53-66)	.983
Female sex, n (%)	67 (41.88%)	19 (43.18%)	48 (41.38%)	.859
Race, n (%)				
White	149 (93.13%)	39 (88.64%)	110 (94.92%)	.176
Black	10 (6.17%)	5 (11.36%)	5 (4.24%)	.137
Other	1 (0.63%)	0 (0%)	1 (0.86%)	1
BMI, kg/m^2, median (IQR)	25.1 (21.4-27.95)	25.45 (22.65-28.28)	25.1 (21-27.9)	.601
Lung allocation score, median (IQR)	36.65 (33.79-41.78)	37.60 (34.74-44.42)	36.050 (33.36-40.13)	.012
Primary diagnosis, n (%)				
Interstitial lung disease	78 (48.75%)	29 (65.91%)	49 (41.24%)	.008
Chronic obstructive pulmonary disease	52 (32.5%)	6 (13.64%)	46 (39.66%)	.001
Cystic fibrosis	19 (11.88%)	4 (9.09%)	15 (12.93%)	.594
Pulmonary hypertension	3 (1.88%)	2 (4.55%)	1 (0.86%)	.184
Sarcoidosis	3 (1.88%)	1 (2.27%)	2 (1.72%)	1
Other	5 (3.13%)	2 (4.55%)	3 (2.59%)	.616
Hospitalized pretransplant, n (%)	39 (24.38%)	15 (34.09%)	24 (20.69%)	.099
In the ICU pretransplant, n (%)	6 (3.75%)	3 (6.82%)	3 (2.59%)	.131
Requiring mechanical ventilation pretransplant, n (%)	2 (1.25%)	1 (2.27%)	1 (0.86%)	.476
Requiring ECMO pretransplant, n (%)	1 (0.63%)	0 (0%)	1 (0.86%)	1
Mean PA pressure median (IQR)	27 (22.25-33)	27 (21-32)	27 (23-33)	.903
Donor				
Age, y, median (IQR)	34 (25-49.25)	34.5 (26-47.25)	34 (24.5-50.25)	.903
Female sex, n (%)	72 (45%)	16 (36.36%)	56 (48.28%)	.214
Race, n (%)				
White	119 (74.38%)	34 (77.27%)	85 (73.28%)	.688
Black	37 (23.13%)	9 (20.46%)	28 (24.14%)	.679
Other	4 (2.5%)	1 (2.27%)	3 (2.59%)	1
BMI, kg/m^2, median (IQR)	25.498 (22.95-30.45)	25.13 (22.88-27.97)	25.713 (22.99-30.63)	.364
Cause of death, n (%)				
Cerebrovascular stroke	44 (27.5%)	12 (27.27%)	32 (27.59%)	1
Head trauma	64 (40%)	19 (43.18%)	45 (38.79%)	.718
Anoxia	51 (31.88%)	13 (29.55%)	38 (32.76%)	.85
Other	1 (0.63%)	0 (0%)	1 (0.86%)	1
Cigarette use >20 pack-y, n (%)	19 (11.88%)	5 (11.36%)	14 (12.07%)	1
Cigarette use within past 6 mo, n (%)	24 (15%)	5 (11.36%)	19 (16.38%)	.62
Chest trauma, n (%)	17 (10.63%)	6 (13.64%)	11 (9.48%)	.566
Last PaO_2:FiO_2 ratio, median (IQR)	457 (400-515)	464 (399-548.5)	455 (406.5-508.25)	.497

BMI, body mass index; ECMO, extracorporeal membrane oxygenation; ICU, intensive care unit; IQR, interquartile range; PA, pulmonary artery; PaO_2:FiO_2, ratio of arterial partial pressure of oxygen to fraction of inspired oxygen; PGD, primary graft dysfunction.

was performed and showed these groups were relatively similar (Supplementary Table 1).

3.2. Model development

Accuracy, AUROC, AUPRC, F1 score, recall, PPV, and NPV were significantly different between all 4 imaging-only approaches ($P < .05$ for each; Table 2). Relative to the other imaging-only model performance metrics, approaches 3 and 4 were the 2 best imaging-only methods and when compared, approach 4 yielded significantly better AUROC (0.63 [0.49-0.72] vs 0.57 [0.47-0.7]; $P < .05$), AUPRC (0.48 [0.35-0.6] vs 0.43 [0.32-0.55]; $P < .05$), F1 score (0.4 [0.28-0.5] vs 0.29 [0.18-0.46]; $P < .05$), recall (0.4 [0.33-0.6] vs 0.33 [0.17-0.5]; $P < .05$), and NPV (0.76 [0.71-0.83] vs 0.75 [0.7-0.8]; $P < .05$ (Table 2, Supplementary Table 2). Because approach 4 had the numerically highest AUROC, AUPRC, and PPV and NPV scores, it was selected for fusion with clinical data. Representative CT scans from 2 patients, 1 with a high and 1 with a low predicted probability of severe PGD within 72 hours of transplant, are shown (Fig. 3).

In total, 98 features were extracted from the EHR, and after feature reduction, 13 were included (Table 3). Compared with clinical EHR data alone, late fusion had a significantly better AUROC (0.74 [0.59-0.85] vs 0.64 [0.54-0.72]; $P < .001$), AUPRC (0.61 [0.47-0.72] vs 0.43 [0.35-0.57]; $P < .001$), F1 score (0.54 [0.36-0.62] vs 0.31 [0.22-0.44]; $P < .001$), recall (0.6 [0.4-0.67] vs 0.2 [0.17-0.4]; $P < .001$), PPV (0.5 [0.33-0.6] vs 0.4 [0.25-0.5]; $P < .01$), and NPV (0.82 [0.77 - 0.88] vs 0.75 [0.72 - 0.8]; $P < 0.001$) performance (Table 2, Supplementary Table 3). Late fusion also had significantly better F1 score (0.54 [0.36-0.62] vs 0.44 [0.22-0.53]; $P < .001$), recall (0.6 [0.4-0.67] vs 0.4 [0.17-0.6]; $P < .001$), PPV (0.5 [0.33-0.6] vs 0.43 [0.3-0.56]; $P < .04$), and NPV (0.82 [0.77-0.88] vs 0.78 [0.74-0.83]; $P = .002$) performance compared with joint fusion II, which was the second best-performing integrative method (Table 2, Supplementary Table 3). When LAS was not removed from the ML pipeline, it appeared in 18% of bootstrap iterations (Supplementary Table 4). In addition, it appeared that late fusion yielded the best results with median AUROC, AUPRC, F1 score, and PPV of 0.68 (0.57-0.8), 0.49 (0.4-0.62), 0.44 (0.4-0.55), and 0.43 (0.33-0.52), respectively (Supplementary Table 5).

4. Discussion

In this study, we incorporated donor-lung CT image features with donor and recipient clinical data to help predict the development of severe PGD after lung transplantation using ML models. Although multiple models have been previously developed to assess donor-lung suitability or predict outcomes after lung transplantation,[9,10,14-16] none of these included donor-lung image features, even though donor-lung CT scans are rapidly becoming a cornerstone of clinical decision making for donor-lung acceptability.[33] Our findings suggest that approach 4 yielded the best overall predictive performance. Furthermore,

Table 2
Model performance metrics.

Modeling approach, median (IQR)	Accuracy[a,c]	AUROC[a,b,c]	AUPRC[a,b,c]	F1 score[a,b,c,d]	Recall[a,b,c,d]	PPV[a,c,d]	NPV[a,b,c,d]
Imaging only							
Approach 1	0.5 (0.35-0.66)	0.49 (0.38-0.6)	0.33 (0.27-0.43)	0.33 (0.14-0.42)	0.4 (0.17-0.83)	0.25 (0.12-0.3)	0.7 (0.63-0.75)
Approach 2	0.5 (0.4-0.61)	0.58 (0.47-0.67)	0.42 (0.34-0.53)	**0.41 (0.33-0.48)**	**0.67 (0.48-0.83)**	0.3 (0.25-0.36)	0.75 (0.68-0.86)
Approach 3	**0.65 (0.6-0.75)**	0.57 (0.47-0.7)	0.43 (0.32-0.55)	0.29 (0.18-0.46)	0.33 (0.17-0.5)	0.33 (0.17-0.5)	0.75 (0.7-0.8)
Approach 4	0.65 (0.55-0.7)	**0.63 (0.49-0.72)**	**0.48 (0.35-0.6)**	0.4 (0.28-0.5)	0.4 (0.33-0.6)	**0.36 (0.25-0.45)**	**0.76 (0.71-0.83)**
Fusion (imaging and clinical data)							
Clinical data only	0.7 (0.65-0.75)	0.64 (0.54-0.72)	0.43 (0.35-0.57)	0.31 (0.22-0.44)	0.2 (0.17-0.4)	0.4 (0.25-0.5)	0.75 (0.72-0.8)
Early fusion	0.6 (0.54-0.65)	0.52 (0.43-0.61)	0.36 (0.28-0.44)	0.31 (0.15-0.4)	0.33 (0.17-0.5)	0.25 (0.14-0.38)	0.73 (0.69-0.77)
Joint fusion II	0.7 (0.65-0.75)	0.72 (0.61-0.79)	0.57 (0.45-0.67)	0.44 (0.22-0.53)	0.4 (0.17-0.6)	0.43 (0.3-0.56)	0.78 (0.74-0.83)
Late fusion	**0.7 (0.65-0.8)**	**0.74 (0.59-0.85)**	**0.61 (0.47-0.72)**	**0.54 (0.36-0.62)**	**0.6 (0.4-0.67)**	**0.5 (0.33-0.6)**	**0.82 (0.77-0.88)**

Clinical Data Only serves as a reference baseline. Fusion approaches were based on Approach 4.
AUPRC, area under the precision-recall curve; AUROC, area under the characteristic receiver operating curve; F1 score, harmonic mean of precision and recall; IQR, interquartile range; NPV, negative predictive value; PPV, positive predictive value.
Bold notation indicates the best performance metric across either imaging-only or fusion-based approaches.

[a] Comparison across all imaging-only approaches performed using Kruskal–Wallis 1-way analysis of variance over 100 bootstrap iterations with statistical significance ($P < .05$). Imaging and Fusion metrics assessed independently.

[b] Statistically significant ($P < .05$) metric comparison between approach 3 and approach 4 over 100 bootstrap iterations performed using Kruskal–Wallis 1-way analysis of variance.

[c] Comparison across all fusion-only approaches performed using Kruskal–Wallis 1-way analysis of variance over 100 bootstrap iterations with statistical significance ($P < .05$). Imaging and Fusion metrics assessed independently.

[d] Statistically significant ($P < .05$) metric comparison between Joint Fusion II and Late Fusion over 100 bootstrap iterations performed using Kruskal–Wallis 1-way analysis of variance.

American Journal of Transplantation 25 (2025) 1339–1349

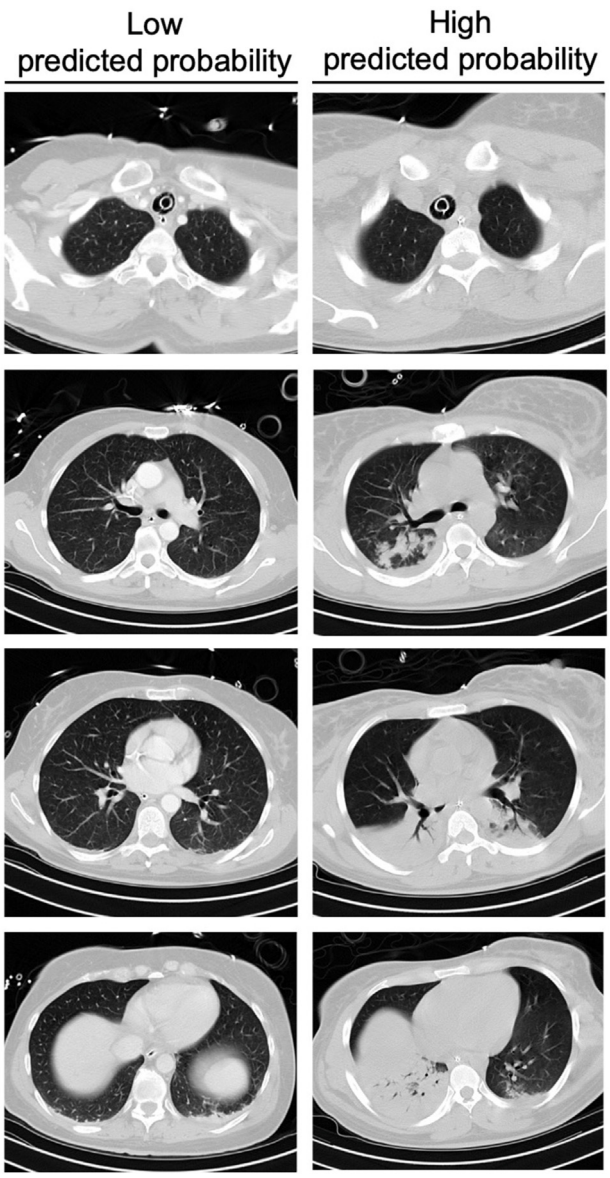

Figure 3. Sample computed tomography scans predicted to have either a low or high probability of grade 3 primary graft dysfunction at 72 hours after bilateral lung transplantation using Approach 4.

fusing donor-lung CT image features with clinical phenotype data enhanced the model's predictive performance compared to a model based solely on clinical data, and late fusion yielded the best performance. This is consistent with changes in clinical practice that led to the widespread use of CT imaging to better assess donor-lung offers. Although the model's predictive performance was modest, these findings illustrate the feasibility of combining CT images with donor and recipient clinical characteristics to improve predictive performance incrementally. Furthermore, these added performance improvements suggest elements within donor-lung CT scans may modulate the likelihood that a recipient will develop severe PGD, and although risk assessment may be straightforward for donor lungs with minimal or severe irregularities, most donor lungs likely exhibit intermediate degrees of pathological findings. In these instances, a validated computational tool could offer the greatest advantage.

Although the best-performing fusion model only achieved a median AUROC of 0.74 (0.59-0.85), this multimodal approach to predictive modeling generated c-statistics consistent with prior studies that relied on significantly larger patient populations and intraoperative factors known to be associated with PGD (eg, cardiopulmonary bypass) to achieve comparable performance.[8,9,16,34] Similarly, by using both imaging and clinical data known only at the time of donor offer-acceptance, this modeling approach not only resembles the clinical decision-making process but is also compatible with the constraints required to be a real-time decision support tool. Moreover, the best fusion approach generated a relatively high median AUPRC (0.61 [0.47-0.72]) and PPV (0.5 [0.33-0.6]), suggesting that despite a modest c-statistic, the predictive output is reliable in predicting severe PGD.

Accurate prediction of severe PGD using donor-lung CT images could enhance donor-recipient matching and potentially reduce the risk of persistent respiratory failure after lung transplantation. While several donor, recipient, and surgical risk factors for severe PGD are widely recognized, their cumulative effects are complex, and a combination of factors may have additive or synergistic effects that make the overall risk for transplantation prohibitive.[30] In addition, multiple studies have demonstrated acceptable posttransplant outcomes without an increased incidence of severe PGD using extended criteria donors with impaired gas exchange, advanced age, or a long smoking history.[35,36] However, clinicians remain hesitant to accept donor lungs that are perceived to be of lower quality. Therefore, identifying methods that better integrate and estimate the cumulative effect of multiple clinical variables on the risk for PGD may not only improve the donor-recipient matching process but also facilitate surgical planning and preparations for postoperative support. Moreover, as new techniques are deployed, new and different risk factors for PGD may emerge, which will require further evaluation.

At present, the current methodologies remain exploratory, and the devised approaches only provide a binary output that attempts to predict the presence or absence of severe PGD after lung transplantation. A more desirable output may be the absolute risk of PGD given a set of donor and recipient characteristics. While we have shown this is technically feasible in prior studies,[16] it will require a larger, more heterogenous population to develop a well calibrated and generalizable algorithm as well as unique infrastructure that can automatically review both discrete data from the EHR and donor-lung CT scan information.

This study has several limitations. First, all included donors were accepted for transplantation, so this study does not address donor suitability. However, it establishes a platform to examine donor suitability specifically for a given recipient in future studies. Second, this was a single-center cohort. A multicenter cohort would enhance the model's generalizability, ensuring that the predictions are applicable across diverse settings. Third, the sample size was small, with only 44 recipients having developed severe PGD. This makes the estimate of error in the model comparatively large, and as larger datasets become available, better measures of performance and degrees of error will be more precise, and we expect models based on a larger cohort would likely perform even better. Fourth, patients were only eligible for

Table 3
Clinical phenotype data selection.

Feature	No. iterations for which feature was selected	Feature	No. iterations for which feature was selected
Recipient not hospitalized before transplant	49	Last donor pH	2
Recipient primary diagnosis - chronic obstructive pulmonary disease	46	Change in donor $PaCO_2$/pH (last − first)	2
Donor hematocrit	32	Donor heavy alcohol use (heavy=2+ drinks/d)	2
Recipient primary diagnosis - interstitial lung disease	27	Time from recipient admission to transplant	2
Female donor - male recipient	26	Time from donor admission to transplant	2
Donor WBC	25	Donor $PaCO_2$	2
Recipient race - Black	23	Recipient on ECMO at time of transplant	2
Donor no. of transfusions during this (terminal) hospitalization	15	Donor cocaine use	2
Recipient primary diagnosis of pulmonary hypertension	15	Donor race - White	2
Donor sex - female	15	Calculated candidate BMI at removal/current time	2
Recipient admitted to hospital at time of transplant	11	Donor A-a gradient rate of change (last − first)	1
Last donor oxygen index	10	Donor race - Other	1
Length of wait	10	Recipient RA pressure	1
Recipient O_2 flow >10 L/min	8	Donor cause of death - cerebrovascular/stroke	1
First donor A-a gradient	7	Male donor - female recipient	1
Recipient race - White	7	First donor tidal volume	1
Donor height - recipient height	6	First donor tidal volume/Ideal body weight	1
Recipient primary diagnosis - other	5	Donor race - Black	1
Recipient PA mean pressure	5	Last donor PEEP	1
First donor PaO_2	4	First donor pH	1
Change in donor PF (first − last)	4	Recipient chronic steroid use at time of transplant	1
Last donor tidal volume/ideal body weight	4	Recipient FVC percent predicted	1
Recipient race - other	3	Donor length of wait	1
Last donor PaO_2	3	Donor height	1
Last donor $PaCO_2$/pH	3	Abnormal donor chest x-ray	1
Recipient - donor sex mismatch	3	Donor cigarette use within last 6 mo	1
First donor PF ratio	3	Donor chest trauma	1
First donor PEEP	2	Donor BMI	1
Donor IV drug use	2	Recipient O_2 requirement at rest before transplant	1

Ninety-eight clinical features were extracted and aggregated from the institutional electronic health record, the United Network for Organ Sharing and the Donor U-Net portal. A least absolute shrinkage and selection operator using Akaike's information criterion was performed on these data 100 times using Akaike's information criterion to reduce the number of clinical features used in the prediction model. Features identified as important in each iteration tallied. Features present in at least 10% of the time were used for predictive modeling.
A-a, alveolar arterial; BMI, body mass index; FVC, forced vital capacity; IV, intravenous; ECMO, extracorporeal membrane oxygenation; PA, pulmonary artery; $PaCO_2$, arterial partial pressure of oxygen, PF, PaO_2:FiO_2, ratio of arterial partial pressure of oxygen to fraction of inspired oxygen; PEEP, positive end expiratory pressure; RA, right atrial; WBC, white blood cell count.

inclusion if transplanted between 2010 and 2019, during which time CT-based donor-lung assessments became routine. This timing may have introduced bias, as patients enrolled earlier likely underwent CT scans for specific clinical indications rather than routine assessment. Furthermore, our ability to retrieve donor CT scans from the study's earlier years was limited, given that the original cohort included patients transplanted at our center starting in 2010. Fifth, the small, single-center population and retrospective study design preclude the use of this model in its current form, and larger, prospective studies are needed to better assess the capabilities of multimodal outcome prediction for lung transplant recipients. Sixth, this multimodal approach to predicting severe PGD appears to incrementally advance model performance, but this improvement comes at the cost of reduced transparency, as feature tracebacks are not readily feasible after data fusion. In such circumstances, it becomes challenging to identify the specific imaging, donor, or recipient characteristics that modulate the risk for severe PGD. However, this tradeoff provides the opportunity to explore robust model performance assessments, such as AUPRC and PPV, which may not aid in transparency but in understanding the performance limits of this clinical decision support. Seventh, the small population precludes subcategory analyses to determine the impact of timing and duration (early vs late vs persistent) of PGD on clinical outcomes. Despite these limitations, our findings demonstrate that fusing donor image features with clinical phenotype data is feasible and improves the predictive performance compared to models based solely on clinical phenotype data alone. Future multicenter studies are necessary to develop more generalizable and better predictive models. In addition, assessing the clinical utility of future models as decision-making tools in practice will be necessary.

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Author contributions

W.M., I.O., A.P.M., and R.R.H. had full access to all of the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis. They also contributed substantially to the writing of the manuscript. S.K., A.G., A.M.L., V.P., D.K., A.E.G., R.N., C.A.W., D.E.B., L.H., R.V.G., P.R.O.P., A.S., H.L., K.N., and M.N.G. contributed substantially to the analysis, interpretation, and design of the work. Yixuan Luo contributed to the design, analysis, and interpretation of the work.

Declaration of competing interest

The authors of this manuscript have conflicts of interest to disclose as described by *American Journal of Transplantation*. R Hachem sits on the advisory board for Mallinckrodt, the clinical events adjudication committee for Transmedics, and has received grant funding from Bristol Myers Squibb and publication support from Boehringer Ingelheim. AM Lai is a shareholder of Johnson & Johnson. A Sotiras has equity in TheraPanacea and has recieved compensation to review grant for the BrightFocus Foundation. AP Michelson is a shareholder of Pfizer and Viatris. The other authors have no conflicts of interest to disclose as described by *American Journal of Transplantation*.

Data availability

The data for these studies are not publicly available as they contain protected health information. Further inquiries may be directed to the corresponding author.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT to improve text clarity for a few select sentences. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ajt.2025.01.039.

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